

ABOUT ME

I leverage weight-space learning to develop generalisable frameworks for sequential tasks. My research bridges theoretical insights with intuitive software engineering for physical systems, world models, and large language models.

SKILLS

- Sequence Modelling
- Meta-Learning
- In-Context Learning
- Weight-Space Learning
- Large Language Models
- Autoregression/Attention
- Linear Recurrence
- Diffusion/Flow Matching
- Latent Action Models
- World Models
- Graph Neural Networks
- Distributed Training
- Bayesian Optimisation
- Neural ODE/PDE Solvers
- Differentiable Simulation
- Foundation Models for Science

Interpersonal: Team leadership, cross-functional collaboration across the UK, France, Japan, Canada, and the USA.

Work ethic: Rigour, rapid iteration, autonomy in designing and running large-scale experiments.

TOOLS


Python


C/C++


PyTorch


JAX


SQL


CUDA


Docker


Kubernetes


Git/Linux


Cloud

RESEARCH INTERESTS

- Model Adaptation**
- Test-time training of LLMs
 - Contextual meta-learning
 - In-context learning
 - Gradient-free adaptation
 - OOD generalisation
 - Weight-space learning
- Deep Sequence Modelling**
- Sub-quadratic attention & linear recurrence
 - Long-horizon sequence modelling

ROUSSEL DESMOND NZOYEM

Machine Learning Researcher pioneering efficient and generalisable approaches to sequential learning.

7 Ingoldsby Ave, Manchester, UK

ddrouslab@gmail.com

+44 (0)7878430616

https://ddrous.github.io

rousseau-desmond-nzoyem

github.com/ddrous

0009-0008-2019-9973

Google Scholar



RESEARCH EXPERIENCE

Postdoctoral Researcher Apr 2026 – Present
University of Manchester | Manchester, UK

- Within the Centre for AI Fundamentals, developing novel paradigms for language model reasoning, world model planning, and Bayesian experimental design under the mentorship of Prof. Samuel Kaski.
- Pursuing sub-quadratic alternatives to standard attention for long-context and non-stationary tasks, grounded in weight-space learning approaches developed during my PhD.

Visiting Researcher Nov 2025 – Jan 2026
AMII, University of Alberta | Edmonton, CA

- Accelerated in-context learning for diffusion and autoregressive sequence models, prototyping and benchmarking novel recurrent, memory-based architectures. Hosted by Prof. Bahareh Tollooshams.
- Investigated efficient weight-space representations as drop-in replacements for cross-attention in language models, with an eye toward high-capacity context windows.

Teaching Assistant Jan 2022 – Sep 2025
University of Bristol | Bristol, UK

- Supported 200+ students across Intro to AI, HPC, Computer Architecture, and Cloud Computing courses.
- Nominated for the prestigious Bristol Teaching Award for excellent lecture quality and student impact.

Data Science Intern (PhD) Jun 2024 – Sep 2024
SLB (Schlumberger) | Oxford, UK

- Prototyped and scaled Graph Neural Network (GNN) inputs by 10x, enabling production-scale fluid flow reconstruction on massive, densely connected subsurface graphs.
- Implemented novel GNN layers to complement JAX's Jraph, achieving a 2x speedup over PyTorch's PyG.
- Achieved few-shot transfer learning to new graphs, and zero-shot super-resolution to larger graphs.

Summer Project Intern (PhD) May 2022 – Aug 2022
HPC Research Group & Bristol Robotics Lab | Bristol, UK

- Integrated differentiable physics simulators (MuJoCo MJX + NVIDIA WARP) for reinforcement learning and closed-loop simulation environments.
- Accelerated algebraic multigrid solvers using GNNs, with rigorous empirical benchmarking across DGL, PyG, and Jraph frameworks.

R&D Intern (MSc) Feb 2021 – Jul 2021
Jacques-Louis Lions Laboratory (Sorbonne University) | Paris, FR

- Developed physics-based generative models for challenging mass-spring-damper dynamical systems requiring robust functional analysis and precise feature engineering.

EDUCATION

PhD in Machine Learning (Interactive AI) Sep 2021 – Nov 2025
University of Bristol | Bristol, UK

- Thesis:** "Learning to Learn Sequential Dynamics: Context-Aware Out-of-Distribution Adaptation for Time Series and Physical Systems", concerned with efficient sequence modelling and meta-learning for non-stationary tasks, with application across physics, text, image, and video modalities.
- Supervisors:** Dr. Tom Deakin, Prof. David Barton, Prof. Simon McIntosh-Smith. **Examiners:** Prof. Nathan Kutz, Dr. Gabriel Leivas Oliveira.
- Published 5+ first-author papers at top-tier venues including ICLR, CoLLAs, AAAI, and SuperComputing.

MSc in Applied Mathematics (CSMI) Sep 2019 – Sep 2021
University of Strasbourg | Strasbourg, FR

- Theoretical and practical analysis of PDEs, signal processing, numerical optimisation, and deep learning. Completed with exceptional distinction (FR: 18.1/20, UK: 1st, US: 4.0).

BSc in Mathematics Nov 2017 – Jul 2019
Aix-Marseille University | Marseille, FR

- Strong focus on statistics, algebra, advanced calculus, and numerical analysis; achieved 15.25/20.

- Bayesian optimal experimental design
- World models & latent space planning

Scientific Machine Learning

- LLM-based code generation for simulation
- Differentiable simulation
- Neural emulators for dynamical systems
- Physics-informed neural networks for PDEs

LANGUAGES

English	●●●●●
French	●●●●●
Japanese	●●●●○
Spanish	●●○○○

TEACHING

- **Thesis Supervision (MSc):** Co-supervised two University of Bristol MSc Data Science students on time series imputation projects.
- **Teaching Assistant (MSc):** Intro to AI, HPC, Computer Architecture, Cloud Computing.
- **Teaching Assistant (BSc):** Scientific Computing, Engineering Mathematics.
- **Award:** Nominated, Bristol Teaching Award (2025).

AWARDS & PRIZES

- Career Development Fellowship by Royal Society (Shortlisted, May 2026)
- Financial Assistance by CRM and ICLR (March and May 2025)
- CDT Studentship by UKRI (June 2021)
- MEXT Scholarship by the Japanese Government (September 2016)
- Cameroon President's Excellence Award (June 2015 & June 2016)

HOBBIES

- **Video games:** I design and play platformers
- **Cinema/music:** I compose movie scores
- **Sports:** I play weekly amateur football
- **Travel:** I explore the globe

REFERENCES

- **David Barton** (University of Bristol)
- **Tom Deakin** (GraphCore)
- **Christophe Prud'homme** (University of Strasbourg)

Associate Degree in Mechatronics

Apr 2017 – Jun 2019

Oshima College of Technology | Oshima, JP

- Intensive training focusing on mechanical, electrical, and computer science engineering.

Associate Degree in Computer Science

Jan 2017 – Apr 2019

University of the People | Pasadena, USA

- Theoretical and applied computer science, followed by web and software development projects.

RECENT PROJECTS

NOVA: World Models with Latent Structural Disentanglement

May 2026

Follow-up to ICLR 2026 Research

- Developed the first world models in weight-space, bypassing modern decoder-heavy architectures to achieve background/foreground/motion disentanglement.
- Engineered a novel anchoring strategy to stabilise weight-space model training.

WARP: Weight-Space Linear Recurrent Neural Networks

Jun 2025

Original Research Published at ICLR 2026

- Introduced WARP, a sub-quadratic, gradient-free in-context learner that unifies weight-space learning with linear recurrence for long-context sequence modelling.
- Demonstrated SOTA performance across real-world classification, time series, and dynamical systems, with a physics-informed variant surpassing baselines by over 10×.
- Led a team of researchers across the UK, France, and the USA to develop JAX code and perform experiments.

Fine-Tuning Gemma-3 for Integer and Decimal Arithmetics

Dec 2025

Follow-up to AAAI 2026 Student Abstract

- Investigated the failure modes of Gemma-3 7B when embedding large numbers, replicating the “Transformers need glasses” phenomenon.
- Implemented the LoRA parameter-efficient fine-tuning pipeline on a single RTX 4080 GPU with limited 16GB VRAM, improving the model's accuracy on arithmetic operations by 28%.

PUBLICATIONS & PREPRINTS

Render, Don't Decode: Weight-Space World Models with Latent Structural Disentanglement

arXiv preprint, 2026.

R. D. Nzoyem, M. Comi.

Weight-Space Linear Recurrent Neural Networks

International Conference on Learning Representations (ICLR), 2026.

R. D. Nzoyem, N. Keshtmand, E. C. Fernandez, I. Tsayem, R. S. Rodriguez, D. A. W. Barton, T. Deakin.

Out-of-Support Generalisation via Weight Space Sequence Modelling

CAO Workshop @ ICLR 2026.

R. D. Nzoyem.

Language Models Do Not Embed Numbers Continuously

AAAI Student Abstract & Poster Program, 2026.

A. Davies, R. D. Nzoyem, N. Ajmeri, T. Silva Filho.

Reevaluating Meta-Learning Optimization Algorithms Through Contextual Self-Modulation

Conference on Lifelong Learning Agents (CoLLAs), 2025.

R. D. Nzoyem, D. A. W. Barton, T. Deakin.

Neural Context Flows for Meta-Learning of Dynamical Systems

International Conference on Learning Representations (ICLR), 2025.

R. D. Nzoyem, D. A. W. Barton, T. Deakin.

Towards Foundational Models for Dynamical System Reconstruction (MixER)

SCOPE Workshop @ ICLR, 2025.

R. D. Nzoyem, G. Stevens, A. Sahota, D. A. W. Barton, T. Deakin.

A comparison of mesh-free differentiable programming [...] for optimal control under PDE constraints

SuperComputing (SC) Workshop on AI4S, 2023. Open-Source Code.

R. D. Nzoyem, D. A. W. Barton, T. Deakin.

OUTREACH, VOLUNTEERING & SERVICES

- **Outreach Ambassador & Tutor:** University of Bristol (Sep 2022 – Sep 2025). Led the *CodeMakers* initiative to inspire young students through after-school programming; featured on the Engineering Includes Me wall.
- **Private Mathematics Instructor:** Complétude (Sep 2019 – Jan 2021) and ExamStar (Sep 2022 – Jul 2024). Affordable lessons for primary and secondary school pupils.
- **Top Reviewer:** International Conference on Machine Learning (ICML), 2025.
- **Reviewer:** ICLR (2025, 2026), NeurIPS (2025), SCOPE@ICLR (2025), CAO@ICLR (2026), Transactions on Machine Learning Research (TMLR, 2025–Present), Journal of Artificial Intelligence Research (JAIR, 2025–Present), Euro-Par (2024).
- **Conference Organiser:** Co-organised the Joint UKRI CDT Conference in AI, ML & Advanced Computing (Jun 2025), and the Interactive AI Spring Research Conference (Mar 2024).